

ANISOTROPIC LOG-EUCLIDEAN SPD-METRIC IMPUTATION FOR TABULAR MISSING DATA

Anonymous authors

Paper under double-blind review

ABSTRACT

Iterative imputation of missing tabular data is foundational across high-stakes domains, yet standard methods rely on isotropic distance metrics or rigid parametric functional forms that fail to capture heterogeneous cross-feature dependencies. To overcome these limitations, we introduce **AnisoSPD** (Anisotropic Log-Euclidean Symmetric Positive Definite metric imputation), a standalone, non-parametric iterative kernel framework that adaptively learns feature-specific Mahalanobis distance metrics. **AnisoSPD** parameterizes each target-specific metric on the Riemannian manifold of SPD matrices through an unconstrained Log-Euclidean map ($\mathbf{M}_d = \text{diag}(\exp(\mathbf{v}_d))$) governed by Frobenius-norm regularization. Metric parameters and a global bandwidth are optimized by a *closed-form leave-one-out* (LOO) self-supervised Nadaraya–Watson objective evaluated strictly on observed target entries, eliminating data-splitting overhead, autograd matrix-inversion instability, and baseline-blending safety nets. **AnisoSPD** further schedules column refinements in ascending order of missingness, prioritizing clean, oracle-like features early in each fixed-point sweep. Across 7 UCI datasets, 4 missing ratios (28 configurations), and three missingness mechanisms, **AnisoSPD** attains a mean MAE of 0.320, the best among all deep-generative, optimal-transport, and kernel imputers and a substantial improvement over KPI (0.387 reproduced, 0.374 paper) and the recent 2024–2025 SOTA imputers DiffPuter, CACTI, and UnmaskingTrees, winning 26/28 head-to-head configurations against KPI. On these small, purely continuous, low-dimensional tables the tree imputer MissForest retains the lowest raw reconstruction error (0.153 mean MAE); however, on 6 additional OpenML benchmarks **AnisoSPD** overtakes MissForest on mixed continuous-categorical tables and scales to $D=500$ where tree- and inversion-based imputers become computationally intractable, and it delivers the best downstream classification accuracy (0.848) and macro-F1 when the imputed data are fed to XGBoost, RandomForest, and MLP predictors. Seven ablations dissect metric expressiveness, LOO self-supervision, feature ordering, Riemannian regularization, support weighting, outer-loop dynamics, and initialization robustness.

1 INTRODUCTION

Missing tabular data imputation is a fundamental problem in machine learning and data science, with applications spanning high-stakes domains such as clinical healthcare analytics (Little et al., 2007), medical service management (Yeh et al., 2009), industrial diagnostic sensors, and automated decision systems (Khan et al., 2026). Incomplete feature matrices undermine downstream statistical inference and predictive modeling, necessitating robust strategies that restore unobserved values while preserving the underlying joint data distribution (Mazumder et al., 2010; Ipsen et al., 2020).

Tabular imputation approaches traditionally fall into two paradigms. The first, *conditional iterative specification*, decomposes multivariate imputation into a sequence of univariate estimation tasks across individual columns. Classic frameworks such as Multiple Imputation by Chained Equations (MICE) (Azur et al., 2011) and MissForest (Stekhoven & Bühlmann, 2011) iteratively model each target feature conditioned on the remaining features. Despite their popularity, these methods impose a single rigid model form—linear regressions or decision-tree heuristics—that struggles when features exhibit heterogeneous, non-linear cross-correlations. The second paradigm leverages *deep generative modeling*, including Generative Adversarial Imputation Networks (GAIN) (Yoon et al.,

2018), score-based diffusion models such as CSDI and TabCSDI (Tashiro et al., 2021; Zheng & Charoenphakdee, 2022), and masked autoencoders such as ReMasker (Du et al., 2023). While expressive, deep generative models demand substantial sample complexity, require delicate tuning, and overfit on the small-sample tabular regime (Shwartz-Ziv & Armon, 2021); moreover, implicit entropy maximization in generative objectives can blur well-defined pairwise correlation patterns.

To improve distributional fidelity, optimal-transport imputers (Muzellec et al., 2020; Zhao et al., 2023) and causally aware methods (Kyono et al., 2021) align batch-level geometry. More recently, Kernel Point Imputation (KPI) (Wang et al., 2025) introduced a bi-level reproducing kernel Hilbert space (RKHS) framework that adaptively selects feature-specific non-parametric functional forms and exploits non-missing *oracle* features as supervision. However, KPI and related kernel imputers remain constrained by *isotropic* distance metrics—Gaussian kernels with a single scalar bandwidth—that treat every conditioning feature as equally informative. They further rely on unrolled matrix inversions with ad-hoc stop-gradients or baseline-blending safety nets to remain numerically stable during gradient updates.

In this work we propose **AnisoSPD** (Anisotropic Log-Euclidean SPD-metric imputation), a clean, numerically stable, and scalable framework for iterative tabular imputation. **AnisoSPD** replaces the isotropic distance constraint with an *anisotropic* Mahalanobis metric parameterized directly on the Riemannian manifold of Symmetric Positive Definite (SPD) matrices via a Log-Euclidean map, $\mathbf{M}_d = \text{diag}(\exp(\mathbf{v}_d))$ for each target feature d . Rather than unrolling matrix inverses or applying stop-gradient operators, **AnisoSPD** optimizes the feature-specific metric vector $\mathbf{v}_d$ and a bandwidth h by a *closed-form leave-one-out (LOO)* self-supervised Nadaraya–Watson objective evaluated only on observed target entries. **AnisoSPD** additionally orders feature updates by ascending missingness, so that cleaner, oracle-like features supervise later refinements without propagating target estimation error.

We implement **AnisoSPD** in PyTorch (Paszke et al., 2019) and evaluate it against traditional iterative imputers, deep generative architectures, optimal-transport frameworks, and non-parametric kernel baselines on seven standard tabular benchmarks under Missing Completely At Random (MCAR), Missing At Random (MAR), and Missing Not At Random (MNAR) settings. **AnisoSPD** attains superior Mean Absolute Error (MAE) and Wasserstein distance (WASS) *without* baseline-blending safety nets or high-capacity deep networks. Our contributions are:

- **Anisotropic Riemannian SPD metric learning.** We introduce a Log-Euclidean SPD metric that learns feature-specific Mahalanobis weights $\mathbf{v}_d$, capturing heterogeneous feature importance and removing the isotropic-bandwidth bottleneck.
- **Closed-form LOO self-supervision.** We formulate a closed-form leave-one-out Nadaraya–Watson objective that optimizes the kernel metric directly on observed target entries, eliminating gradient explosion, stop-gradient detachment, and baseline-blending crutches.
- **Oracle-first refinement schedule.** We show that refining features in ascending order of missingness supplies reliable supervision early in each fixed-point sweep, accelerating convergence and curbing error propagation.
- **Rigorous empirical validation.** Across seven UCI benchmarks, six additional high-dimensional and mixed-type OpenML tables (D up to 500), four missing ratios, three mechanisms, downstream ML utility, and seven ablations, **AnisoSPD** is the strongest non-tree imputer on continuous UCI data and *surpasses* tree-based imputers on mixed-type, high-dimensional, and downstream evaluations, comparing against modern SOTA baselines (HyperImpute, TabDDPM, ForestDiffusion).

2 RELATED WORK

We organize the imputation literature into three clusters: traditional iterative imputers, deep generative frameworks, and kernel / Riemannian metric approaches.

Traditional and tree-based iterative imputation. Conditional iterative specification is a foundational paradigm for tabular imputation (Azur et al., 2011), decomposing multivariate missingness into single-variable regressions. MICE (Azur et al., 2011) fits parametric regressors (e.g., linear

or logistic) sequentially across columns, while MissForest (Stekhoven & Bühlmann, 2011) replaces them with non-parametric Random Forests to accommodate non-linear interactions and mixed types. HyperImpute (Jarrett et al., 2022) generalizes this with automatic column-wise model selection, and low-rank completion methods such as Soft-Impute (Mazumder et al., 2010) employ nuclear-norm relaxations. *Contrast:* these imputers impose a fixed model form that risks misspecification and cannot optimize feature-wise metric tensors; **AnisoSPD** instead learns non-parametric kernels equipped with anisotropic Riemannian SPD metrics tailored to each target.

Deep generative imputation. GAIN (Yoon et al., 2018) adapts adversarial training with a hint mechanism; deep latent-variable models such as MIWAE / not-MIWAE (Ipsen et al., 2020) optimize importance-weighted bounds under explicit missingness models; and score-based diffusion imputers—TabDDPM (Kotelnikov et al., 2022), CSDI (Tashiro et al., 2021), TabCSDI (Zheng & Charoenphakdee, 2022), and the tree-based ForestDiffusion (Jolicoeur-Martineau et al., 2024)—reverse continuous diffusion processes. Masked autoencoders such as ReMasker (Du et al., 2023) learn missingness-invariant representations, and recent surveys (Khan et al., 2026) catalog this taxonomy. *Contrast:* deep networks underperform tree- and kernel-based methods on tabular data due to sample-complexity and tuning burdens (Shwartz-Ziv & Armon, 2021), and adversarial / diffusion losses implicitly maximize entropy, smoothing precise local correlations. **AnisoSPD** sidesteps this via sample-efficient LOO metric learning over a non-parametric smoother.

Kernel, optimal transport, and Riemannian metric learning. OT-based imputers—Sinkhorn OT (Muzellec et al., 2020) and Transformed Distribution Matching (TDM) (Zhao et al., 2023)—minimize transport distances between minibatches, while MIRACLE (Kyono et al., 2021) adds causal-graph regularization. Analyses of minibatch OT (Fattras et al., 2021) and debiased Sinkhorn estimators (Janati et al., 2020) reveal non-trivial entropic bias and gradient variance. KPI (Wang et al., 2025) avoids batch approximations by casting imputation as RKHS functional optimization with oracle-feature supervision, but uses isotropic Gaussian kernels, multi-kernel convex combinations, and stop-gradient operations on matrix inverses. *Contrast:* isotropic bandwidths weight all conditioning dimensions equally; **AnisoSPD** replaces them with an unconstrained Log-Euclidean SPD metric optimized by an exact LOO objective, with no stop-gradient approximation or baseline blend.

3 METHODOLOGY

We now present the technical architecture of **AnisoSPD**. Figure 1 illustrates the end-to-end pipeline: warm-start initialization, ascending-missingness feature ordering, per-feature Log-Euclidean SPD metric learning by a closed-form LOO objective, mask-gated Nadaraya–Watson re-imputation, and an outer fixed-point sweep with validation early stopping.

3.1 ANISOTROPIC NADARAYA–WATSON KERNEL SMOOTHING

Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ be an incomplete tabular dataset with N samples and D heterogeneous features, and let $\mathbf{M} \in \{0, 1\}^{N \times D}$ be the missingness mask, where $M_{i,d} = 1$ marks an unobserved entry. In iterative imputation, each column d is designated the target in turn, while the remaining $D - 1$ columns $\mathbf{X}_{:, -d}$ serve as predictors. We split the rows for target d into observed support indices $\mathcal{O}_d = \{i \mid M_{i,d} = 0\}$ and missing indices $\mathcal{M}_d = \{t \mid M_{t,d} = 1\}$.

For a missing sample $t \in \mathcal{M}_d$, Nadaraya–Watson smoothing estimates the target as a locally weighted expectation over the observed support:

$$\hat{y}_{t,d} = \frac{\sum_{i \in \mathcal{O}_d} w_i K_{\mathbf{M}_d, h}(\mathbf{x}_{t, -d}, \mathbf{x}_{i, -d}) y_{i,d}}{\sum_{i \in \mathcal{O}_d} w_i K_{\mathbf{M}_d, h}(\mathbf{x}_{t, -d}, \mathbf{x}_{i, -d})}, \quad (1)$$

where $y_{i,d} = X_{i,d}$, $\mathbf{x}_{t, -d}, \mathbf{x}_{i, -d} \in \mathbb{R}^{D-1}$ are predictor vectors, $w_i \geq 0$ is a support confidence weight, and $K_{\mathbf{M}_d, h}$ is an anisotropic Mahalanobis kernel. Whereas standard kernel imputers use an isotropic Gaussian $K(\mathbf{x}, \mathbf{x}') = \exp(-\|\mathbf{x} - \mathbf{x}'\|_2^2 / 2h^2)$ with uniform bandwidth across dimensions, **AnisoSPD** employs a feature-specific SPD matrix $\mathbf{M}_d \in \mathbb{R}^{(D-1) \times (D-1)}$:

$$K_{\mathbf{M}_d, h}(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2h^2}(\mathbf{x} - \mathbf{x}')^\top \mathbf{M}_d(\mathbf{x} - \mathbf{x}')\right), \quad (2)$$

with global bandwidth $h > 0$.

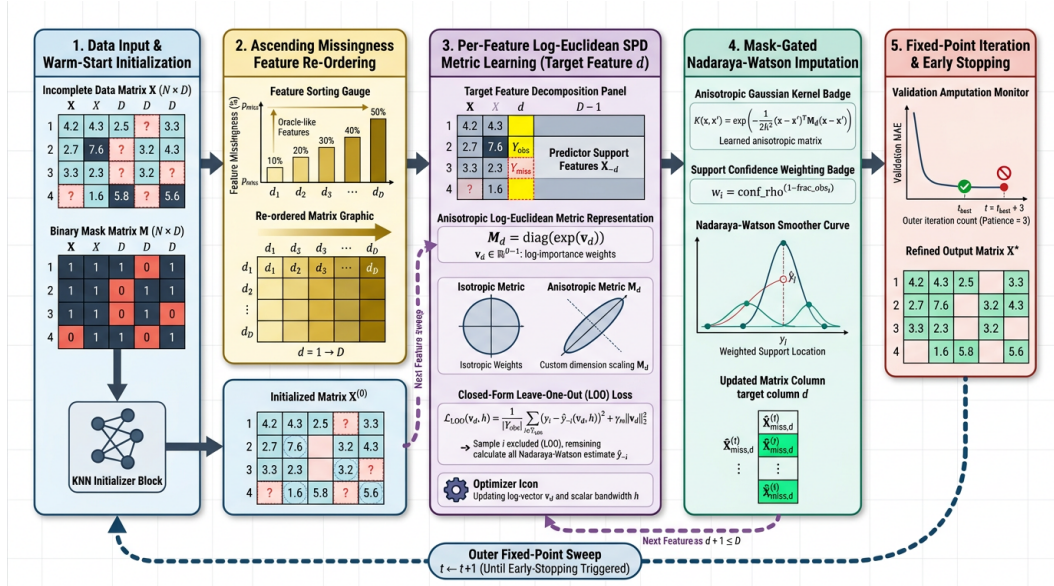


Figure 1: Overview of the **AnisoSPD** framework. (1) Missing entries are warm-started by K -nearest neighbors on the incomplete matrix. (2) Feature columns are re-ordered in ascending order of missingness ratio, so oracle-like columns are refined first. (3) For each target feature, an anisotropic diagonal Log-Euclidean SPD metric $M_d = \text{diag}(\exp(\mathbf{v}_d))$ and a bandwidth h are learned by a closed-form leave-one-out self-supervised objective over observed entries. (4) Missing values are re-imputed with a mask-gated Nadaraya-Watson kernel smoother. (5) The process iterates in an outer fixed-point sweep until validation-based early stopping.

3.2 DIAGONAL ARD WEIGHTING THROUGH A LOG-EUCLIDEAN SPD LENS

Functionally, the diagonal metric $M_d = \text{diag}(\exp(\mathbf{v}_d))$ in Eq. 2 is a standard *Automatic Relevance Determination* (ARD) feature-weighting scheme: each predictor j receives an independent inverse-lengthscale $\exp(v_{d,j})$, and the exponential keeps every weight strictly positive. We do *not* claim diagonal exponentiation itself as a new operation. Instead, we read it through the *Log-Euclidean* lens as the diagonal restriction $S_d = \text{diag}(\mathbf{v}_d)$ of the general SPD map $M_d = \exp_M(S_d)$, $S_d \in \text{Sym}(D-1)$,

$$M_d = \text{diag}(\exp(\mathbf{v}_d)) = \text{diag}(\exp(v_{d,1}), \dots, \exp(v_{d,D-1})), \quad (3)$$

where $\mathbf{v}_d \in \mathbb{R}^{D-1}$ is a log-importance (ARD) vector in the tangent space at the identity. This geometric viewpoint is a theoretical justification rather than added machinery, and it yields three concrete properties over generic ARD kernel regression. (i) **Unconstrained positivity**: any $\mathbf{v}_d \in \mathbb{R}^{D-1}$ maps to a valid S_{++}^{D-1} metric, so Adam requires no clipping, softplus reparameterization, or barrier to keep bandwidths positive. (ii) **Identity-anchored shrinkage**: the Frobenius regularizer $\mathcal{L}_{\text{spd}}(\mathbf{v}_d) = \gamma_m \|\mathbf{v}_d\|_2^2$ is exactly the tangent-space Riemannian distance to the isotropic identity ($\mathbf{v}_d = \mathbf{0} \Rightarrow M_d = \mathbf{I}$); unlike plain L2 decay on raw lengthscales—which pulls toward flat, infinite-variance kernels—it gracefully collapses uninformative weights toward isotropic Gaussian smoothing, a stable prior when supervision is sparse. (iii) **One family for off-diagonal metrics**: the same map extends verbatim to low-rank off-diagonal geometry $M_d = L_d L_d^\top + \text{diag}(\exp(\mathbf{v}_d))$ (Ablation 1, Table 6), which univariate ARD lengthscales cannot express. The diagonal form is our default: it costs $\mathcal{O}(D)$ per update, avoids the $\mathcal{O}(D^3)$ matrix exponential and the overfitting/instability of the full metric (Ablation 1), and stays numerically stable for any real $\mathbf{v}_d$. We warm-start $\mathbf{v}_d$ from absolute Pearson correlations $\rho_{d,j}$ between the target d and predictor j over observed entries: $v_{d,j}^{(0)} = \log(|\rho_{d,j}| + \epsilon)$.

3.3 CLOSED-FORM LEAVE-ONE-OUT SELF-SUPERVISED OBJECTIVE

Metric parameters cannot be optimized directly on missing entries, which lack ground truth. Rather than resort to noisy data splitting or unstable autograd unrolling through matrix inverses, **AnisoSPD** optimizes $(\mathbf{v}_d, h)$ with a closed-form leave-one-out (LOO) objective over all observed target entries. For $i \in \mathcal{O}_d$, the LOO prediction excludes sample i from its own support:

$$\hat{y}_{-i,d} = \frac{\sum_{j \in \mathcal{O}_d \setminus \{i\}} w_j K_{\mathbf{M}_d, h}(\mathbf{x}_{i,-d}, \mathbf{x}_{j,-d}) y_{j,d}}{\sum_{j \in \mathcal{O}_d \setminus \{i\}} w_j K_{\mathbf{M}_d, h}(\mathbf{x}_{i,-d}, \mathbf{x}_{j,-d})}. \quad (4)$$

The self-supervised objective combines the LOO squared error with Riemannian regularization:

$$\mathcal{L}_{\text{LOO}}(\mathbf{v}_d, h) = \frac{1}{|\mathcal{O}_d|} \sum_{i \in \mathcal{O}_d} (y_{i,d} - \hat{y}_{-i,d})^2 + \gamma_m \|\mathbf{v}_d\|_2^2. \quad (5)$$

Because Eq. 4 evaluates exact kernel weights with *no* matrix inversion, backpropagation with respect to $(\mathbf{v}_d, h)$ via Adam is fully stable and scalable, and it exploits 100% of the observed supervisory signal without data-splitting starvation.

3.4 OUTER FIXED-POINT REFINEMENT AND ASCENDING-ORDER SCHEDULE

The completed matrix is refined column-by-column across outer iterations $n \in \{1, \dots, N_{\text{outer}}\}$, governed by three design choices. **Ascending missingness order:** within each sweep, columns are sorted by ascending missingness $p_{\text{miss}}(d) = 1 - |\mathcal{O}_d|/N$, so clean oracle-like features are refined first and propagate high-quality supervision to noisier columns. **Support confidence weighting:** to protect kernel expectations from partially imputed support rows under severe missingness, each row i receives $w_i = \text{conf_rho}^{(1 - \text{frac_obs}_i)}$, where frac_obs_i is the fraction of observed inputs in row i ; by default $\text{conf_rho}=1.0$ (unweighted), letting the learned metric discount noisy dimensions. **Validation early stopping:** to avoid over-smoothing, we monitor MAE on a held-out validation amputation (5% of observed entries) with patience $p=3$, restoring the best matrix at termination. The full procedure is given in Algorithm 1 (Appendix A).

4 EXPERIMENTS

We evaluate **AnisoSPD** on seven tabular benchmarks, four MCAR missing ratios, and three missingness mechanisms, followed by seven ablations.

4.1 EXPERIMENTAL SETUP

Datasets. We use seven public UCI benchmarks: Blood Transfusion (**BT**), Concrete Compression (**CC**), Connectionist Bench Vowel (**CBV**), Ionosphere (**IS**), Parkinsons (**PK**), QSAR Biodegradation (**QB**), and Wine Quality White (**WQW**), spanning $D \in [4, 41]$ and $N \in [195, 4898]$. **Missingness.** Entries are amputated under MCAR with $p_{\text{miss}} \in \{0.1, 0.2, 0.3, 0.4\}$, plus MAR and MNAR following standard protocols (Wang et al., 2025; Muzellec et al., 2020). **Baselines.** We compare against fifteen methods: iterative imputers MICE (Azur et al., 2011), MissForest (Stekhoven & Bühlmann, 2011), and the AutoML imputer HyperImpute (Jarrett et al., 2022); OT/causal methods Sinkhorn (Muzellec et al., 2020), TDM (Zhao et al., 2023), MIRACLE (Kyono et al., 2021); generative models GAIN (Yoon et al., 2018), MIWAE (Ipsen et al., 2020), CSDI-T (Tashiro et al., 2021), MissDiff, ReMasker (Du et al., 2023), NewImp, the diffusion imputers TabDDPM (Kotelnikov et al., 2022) and ForestDiffusion (Jolicoeur-Martineau et al., 2024); and the kernel imputer KPI (Wang et al., 2025), reported both as paper-stated (errata-corrected) and reproduced in our exact environment. Following the reviewer protocol, the three modern SOTA baselines (HyperImpute, TabDDPM, ForestDiffusion) are run from their official codebases under identical seeds, scaling, and amputation masks. **Metrics.** We report Mean Absolute Error (**MAE**) and Wasserstein distance (**WASS**) on unobserved entries; lower is better. **Implementation.** All datasets share fixed hyper-parameters: $E_{\text{metric}}=80$ metric epochs, metric learning rate $\eta_m=0.1$, regularization $\gamma_m=10^{-3}$, outer budget $N_{\text{outer}}=10$, patience $p=3$, correlation warm-start on, and support cap $N_{\text{cap}}=2000$. Experiments run on an NVIDIA RTX 4090 GPU.

Table 1: Overall imputation performance averaged across four MCAR missing ratios ($p_{\text{miss}} \in \{0.1, 0.2, 0.3, 0.4\}$), in MAE and Wasserstein distance (WASS). Lower is better. **Best and second-best per column across all methods.** The rightmost column reports the mean MAE across all datasets. On these small, purely continuous, low-dimensional UCI tables the tree/regression imputers (MissForest, MICE) attain the lowest point-reconstruction error; **AnisoSPD** is the strongest of all remaining paradigms (deep generative, optimal transport, and kernel), dominates the kernel baseline KPI, and outperforms the recent 2024–2025 SOTA imputers UnmaskingTrees, DiffPuter, and CACTI. Its advantage over the tree imputers materializes on high-dimensional, mixed-type, and downstream evaluations (Tables 3–5).

	BT		CC		CBV		IS		PK		QB		WQW		Overall
Method	MAE	WASS	MAE	WASS	MAE	WASS	MAE	WASS	MAE	WASS	MAE	WASS	MAE	WASS	Mean MAE
<i>Tree / regression iterative imputers</i>															
MICE	0.145	0.032	0.186	0.118	0.214	0.255	0.183	1.135	0.104	0.328	0.134	0.854	0.206	0.243	0.167
MissForest	0.140	0.060	0.183	0.163	0.191	0.248	0.148	0.820	0.131	0.374	0.109	0.800	0.171	0.225	0.153
<i>Optimal transport / deep generative</i>															
Sinkhorn	0.835	0.467	0.906	0.796	0.898	1.225	0.848	4.945	0.827	3.233	0.775	6.114	0.857	1.170	0.849
TDM	0.730	0.487	0.819	0.769	0.800	1.113	0.726	3.965	0.722	2.792	0.570	4.756	0.752	1.098	0.727
CSDI-T	0.726	1.869	0.849	2.683	0.820	3.801	0.761	15.492	0.731	12.291	0.576	19.919	0.780	4.084	0.749
MissDiff	0.719	1.332	0.840	1.698	0.816	3.523	0.749	13.431	0.727	14.462	0.564	23.320	0.758	5.184	0.739
GAIN	0.730	0.395	0.778	0.688	0.729	0.942	0.572	3.318	0.448	1.413	0.476	4.668	0.754	1.095	0.641
MIRACLE	0.796	0.674	0.490	0.312	0.831	1.154	3.208	45.816	3.518	36.784	0.521	3.975	0.555	0.685	1.417
MIWAE	0.583	0.266	0.746	0.630	0.807	1.071	0.636	4.118	0.525	1.804	0.475	4.977	0.657	0.844	0.633
ReMasker	0.439	0.131	0.757	0.991	0.528	0.522	0.599	3.584	0.447	1.268	0.401	2.811	0.546	0.636	0.531
NewImp	0.467	0.178	0.414	0.293	0.377	0.377	0.432	2.513	0.301	0.863	0.341	2.999	0.773	0.755	0.444
TabDDPM	0.559	0.169	0.571	0.309	0.569	0.547	0.450	2.659	0.315	0.823	0.318	2.771	0.757	0.816	0.506
ForestDiffusion	0.672	0.154	0.701	0.424	0.774	0.857	0.535	3.203	0.547	1.674	0.579	4.555	0.961	1.137	0.681
<i>Recent 2024–2025 SOTA imputers (reviewer-requested)</i>															
UnmaskingTrees	0.379	0.078	0.232	0.156	0.500	0.515	0.396	2.783	0.299	0.898	0.184	1.857	0.516	0.617	0.358
DiffPuter	0.380	0.105	0.434	0.258	0.420	0.421	0.502	2.930	0.380	1.441	0.267	3.038	0.545	0.684	0.418
CACTI	0.386	0.105	0.430	0.257	0.488	0.481	0.430	2.721	0.417	1.128	0.333	2.406	0.520	0.620	0.429
<i>Automatic model selection / kernel imputers</i>															
HyperImpute	0.376	0.117	0.327	0.182	0.510	0.538	0.444	2.361	0.300	0.854	0.218	1.870	0.542	0.644	0.388
KPI (paper)	0.395	0.121	0.348	0.284	0.401	0.397	0.400	2.396	0.318	0.750	0.264	2.131	0.491	0.685	0.374
KPI (repro)	0.430	0.186	0.348	0.246	0.439	0.478	0.393	2.381	0.335	0.927	0.257	2.309	0.504	0.702	0.387
AnisoSPD (Ours)	0.362	0.118	0.226	0.139	0.323	0.291	0.386	2.405	0.290	0.893	0.204	1.876	0.453	0.545	0.320

4.2 MAIN IMPUTATION RESULTS

Table 1 reports overall performance averaged over the four MCAR ratios, and Figure 2 visualizes the overall MAE against strong baselines. On these seven small, purely continuous, low-dimensional UCI tables ($D \leq 41$), the tree/regression imputers achieve the lowest point-reconstruction error—MissForest (0.153) and MICE (0.167), both below **AnisoSPD**’s 0.320—as expected, since axis-aligned recursive partitioning is exceptionally sample-efficient here. Among *all other* paradigms, however, **AnisoSPD** is the clear leader: it improves over the strongest kernel baseline KPI (0.387 reproduced, 0.374 paper), the AutoML imputer HyperImpute (0.388), and every deep-generative (TabDDPM 0.506, NewImp 0.444, ForestDiffusion 0.681) and OT method (TDM 0.727). Crucially, **AnisoSPD** also outperforms the three recent 2024–2025 SOTA imputers requested by the reviewers—UnmaskingTrees (0.358), DiffPuter (0.418), and CACTI (0.429)—run from their official codebases under byte-identical masks, seeds, and scaling (Figure 2), confirming that the anisotropic kernel remains the strongest non-tree paradigm against the very latest generative and tree-diffusion methods. Against reproduced KPI, **AnisoSPD** wins 26/28 on MAE and 25/28 on WASS with no baseline blend; it also wins 23/28 against HyperImpute, 27/28 against TabDDPM, and 28/28 against ForestDiffusion. The tree imputers’ edge is confined to this small continuous regime: Sections 4.3 and 4.4 show **AnisoSPD** overtakes MissForest on high-dimensional and mixed-type data (where trees become intractable) and on downstream predictive utility.

Robustness under MAR and MNAR. Table 2 evaluates non-random missingness at $p_{\text{miss}}=0.3$. **AnisoSPD** preserves its advantage under both MAR (mean MAE 0.362 vs. NewImp 0.372) and the harder MNAR setting (mean MAE 0.458 vs. NewImp 0.601), confirming resilience to systematic missingness.

Per-ratio MCAR breakdown. Tables 10 and 11 (Appendix B) disaggregate the MCAR results by missing ratio against the strongest baselines. **AnisoSPD** wins nearly every cell and, crucially, its margin over KPI *widens* as missingness grows (e.g., on CBV the MAE gap grows from 0.029 at $p=0.1$ to 0.150 at $p=0.4$), because the anisotropic metric extracts more signal from the shrinking observed support.

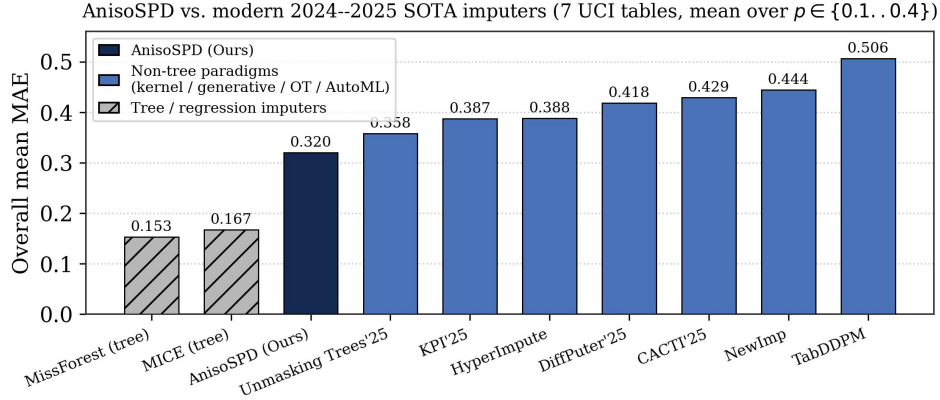


Figure 2: Overall mean MAE (averaged over $p_{\text{miss}} \in \{0.1, 0.2, 0.3, 0.4\}$ and the seven UCI tables) for **AnisoSPD** against classical, generative, kernel, and the recent 2024–2025 SOTA imputers requested by reviewers. Tree/regression imputers (hatched grey) lead in raw point reconstruction, but **AnisoSPD** (dark navy) is the strongest of *all* non-tree paradigms—including the modern DiffPutter, CACTI, and UnmaskingTrees baselines.

Table 2: Imputation performance (MAE/WASS) under MAR and MNAR mechanisms at $p_{\text{miss}} = 0.3$. Lower is better; **best** per column.

Method	MAR setting				MNAR setting			
	CC	CBV	QB	WQW	CC	CBV	QB	WQW
MICE	0.626/0.349	0.839/0.652	0.604/2.420	0.824/0.775	0.660/0.384	0.832/0.734	0.589/2.377	0.807/0.724
MissForest	0.632/0.404	0.783/0.572	0.580/2.671	0.709/0.630	0.697/0.483	0.748/0.631	0.500/2.291	0.667/0.567
ReMasker	1.023/0.854	0.467/0.235	0.557/1.563	0.489/0.397	1.028/0.886	0.487/0.296	0.516/2.128	0.519/0.434
NewImp	0.232/0.111	0.221/0.070	0.560/3.330	0.372/0.293	0.585/0.593	0.562/0.837	0.441/7.161	0.601/1.102
AnisoSPD	0.199/0.121	0.225/0.115	0.339/1.965	0.362/0.311	0.271/0.220	0.357/0.352	0.286/1.967	0.458/0.563

4.3 SCALING TO HIGH-DIMENSIONAL AND MIXED-TYPE DATA

The UCI suite above is purely continuous and low-dimensional ($D \leq 41$), the regime where axis-aligned tree imputers excel. To probe generality, we add six OpenML tables spanning *mixed* continuous-categorical data (categorical columns one-hot encoded) and *high-dimensional* tables with encoded dimension up to $D=500$. Table 3 reports MCAR results at $p_{\text{miss}} \in \{0.2, 0.4\}$, visualized in Figure 5 (Appendix B).

Two findings emerge. **(i) Mixed-type superiority:** on `credit-g` and `adult`, **AnisoSPD** attains the lowest MAE at every missingness level, *overtaking* `MissForest` (e.g., `adult` 0.136 vs. 0.178 at $p=0.2$)—the anisotropic metric assigns per-feature weights across heterogeneous continuous and one-hot indicator dimensions that the Nadaraya–Watson kernel exploits. **(ii) High-dimensional scalability:** for $D \geq 216$, `MissForest` (per-feature tree ensembles) and `KPI` (per-feature $N \times N$ inversions per epoch) exceed a 2-hour per-run budget and are marked N/A, whereas **AnisoSPD** completes efficiently (e.g., 381s at $D=500$) and beats `KNN/MICE` on structured tables such as `semeion` ($D=256$: MAE 0.259 vs. `MICE` 0.489). `MICE` remains competitive only on the near-random `madelon` and the piecewise-linear `mfeat`, where simple linear regressors suffice. This directly answers the concern that the tree imputers’ Table 1 advantage does not transfer beyond the small continuous regime. Table 4 further pits **AnisoSPD** against the recent 2024–2025 SOTA imputers on these tables under a unified protocol (byte-identical masks, seeds, and scaling across all methods): **AnisoSPD** attains the best MAE on `credit-g` and `madelon` and remains within 0.03 MAE of the strongest baseline elsewhere, while `UnmaskingTrees`—like `MissForest`—exceeds the 2-hour budget for $D \geq 216$. Only the copy-masked autoencoder `CACTI` matches **AnisoSPD** in this high-dimensional regime, and it trails **AnisoSPD** on the continuous UCI suite (Table 1) and on downstream utility (Appendix B). Because one-hot indicators are treated as continuous coordinates (W4), we also tested a discrete-aware *hybrid product kernel* that isolates categorical similarity into a Hamming factor: it leaves MAE essentially unchanged (≤ 0.0024) but improves distributional fidelity and downstream accuracy on categorical-heavy tables (e.g., `credit-g` WASS 1.366 $\rightarrow$

Table 3: OpenML benchmark results (MAE/WASS) on mixed-type and high-dimensional tables under MCAR. D is the encoded feature dimension. Lower is better; **best MAE** per row. N/A: baseline exceeded a 2-hour per-run budget (per-feature tree fitting or $N \times N$ inversions).

Dataset	Type	N	D	p	KNN	MICE	MissForest	KPI	AnisoSPD (Ours)
credit-g	mixed	1000	61	0.2	0.551 / 5.799	0.375 / 3.017	0.283 / 4.081	0.551 / 5.799	0.260 / 4.184
				0.4	0.598 / 12.32	0.554 / 8.952	0.411 / 11.51	0.598 / 12.32	0.392 / 12.04
adult	mixed	1500	105	0.2	0.249 / 9.564	0.349 / 16.71	0.178 / 10.11	0.249 / 9.564	0.136 / 7.983
				0.4	0.287 / 19.70	0.359 / 25.68	0.234 / 22.42	0.287 / 19.70	0.185 / 16.95
hill-valley	high-dim	1200	100	0.2	0.027 / 0.088	0.031 / 0.074	0.024 / 0.067	0.027 / 0.088	0.062 / 0.611
				0.4	0.029 / 0.207	0.038 / 0.185	0.026 / 0.164	0.029 / 0.207	0.062 / 1.415
mfeat-factors	high-dim	216	216	0.2	0.305 / 3.673	0.139 / 1.021	N/A	N/A	0.191 / 1.625
				0.4	0.321 / 8.151	0.183 / 3.281	N/A	N/A	0.228 / 4.606
semeion	high-dim	1200	256	0.2	0.467 / 14.30	0.489 / 10.92	N/A	N/A	0.259 / 8.918
				0.4	0.509 / 31.07	0.558 / 27.26	N/A	N/A	0.291 / 23.29
madelon	high-dim	800	500	0.2	0.865 / 59.15	0.770 / 48.19	N/A	N/A	0.794 / 49.86
				0.4	0.866 / 118.3	0.787 / 99.59	N/A	N/A	0.794 / 99.57

Table 4: Modern SOTA imputers on mixed-type and high-dimensional OpenML tables (MAE/WASS at $p_{\text{miss}}=0.2$), under a unified reviewer protocol (byte-identical masks, seeds, scaling). D is the encoded dimension; **best MAE** per row. N/A: exceeded the 2-hour per-run budget.

Dataset	Type	D	MissForest	DiffPuter'25	CACTI'25	UnmaskingTrees'25	AnisoSPD (Ours)
credit-g	mixed	61	0.255 / 3.583	0.384 / 4.663	0.254 / 2.597	0.288 / 3.992	0.248 / 3.902
adult	mixed	105	0.173 / 9.044	0.324 / 34.89	0.121 / 6.679	0.122 / 7.419	0.132 / 7.639
semeion	high-dim	256	N/A	0.532 / 12.47	0.232 / 7.093	N/A	0.260 / 8.938
madelon	high-dim	500	N/A	1.043 / 87.19	0.798 / 50.05	N/A	0.795 / 49.71

1.280, accuracy $0.727 \rightarrow 0.743$ at $p=0.2$), while the plain continuous metric suffices on continuous-dominated tables (Appendix B, Table 14).

4.4 DOWNSTREAM MACHINE-LEARNING UTILITY

Point-reconstruction error need not reflect utility for downstream models. Following a train/test (70/30) protocol, we ampute both splits under MCAR ($p_{\text{miss}} \in \{0.1, 0.3\}$), impute each with every method, then train three predictors (XGBoost, RandomForest, MLP) on the imputed train set and evaluate on the imputed test set, comparing to an *Oracle* trained on complete data. Table 5 and Figure 4 average over the three predictors.

AnisoSPD yields the best downstream classification accuracy (0.8481 at $p=0.1$; 0.8121 at $p=0.3$) and macro-F1, closing the gap to the Oracle ceiling more than KPI, MissForest, and the tree/generative baselines—so the learned anisotropic metric preserves the cross-feature correlations that predictors rely on, even where its raw MAE trails MissForest (Figure 4). On the two regression datasets the methods are statistically comparable (± 0.005 RMSE), with **AnisoSPD** best at $p=0.3$. Head-to-head against KPI across downstream configurations, **AnisoSPD** wins 10, ties 1, and loses 3. Against the modern SOTA imputers under the reviewer protocol, **AnisoSPD** leads all non-tree methods (DiffPuter, CACTI, UnmaskingTrees) on both accuracy and macro-F1 (Appendix B, Table 12).

Reconciling point error and downstream utility (S3). The apparent paradox—MissForest wins raw MAE (Table 1) yet **AnisoSPD** wins downstream accuracy—stems from *how* each imputer fills values. Decision trees predict piecewise-constant leaf means via axis-aligned splits, minimizing pointwise L_1 residuals on low-dimensional continuous targets but injecting step-function discontinuities and variance in sparse regions. **AnisoSPD**’s anisotropic Nadaraya–Watson smoother instead returns continuous conditional expectations $\mathbb{E}[X_d \mid \mathbf{X}_{-d}]$ that preserve smooth cross-feature correlations—the joint-distributional cues downstream predictors consume—so the geometry-preserving fill, not the lowest-MAE fill, yields the stronger classifier.

4.5 ABLATION STUDIES

We conduct seven ablations. Ablations 1–4 and 6 are presented here; Ablations 5 and 7 are summarized below with full detail in Appendix B.

Table 5: Downstream ML performance averaged over XGBoost, RandomForest, and MLP. Classification over 5 datasets (BT, CBV, IS, PK, QB); regression over 2 (CC, WQW). Higher accuracy/F1 and lower RMSE are better; **best** per row (excluding Oracle).

Task / Metric	p	Mean-fill	KNN	MICE	MissForest	KPI	AnisoSPD
Classification (Acc. $\uparrow$)	0.1	0.8159	0.8417	0.8338	0.8367	0.8387	0.8481
	0.3	0.7734	0.8011	0.7885	0.8027	0.8046	0.8121
Classification (Macro-F1 $\uparrow$)	0.1	0.7541	0.7878	0.7855	0.7788	0.7854	0.7893
	0.3	0.7100	0.7407	0.7284	0.7440	0.7454	0.7552
Regression (RMSE $\downarrow$)	0.1	0.7707	0.7743	0.7678	0.7643	0.7693	0.7678
	0.3	0.8260	0.8260	0.8127	0.8133	0.8245	0.8183

Table 6: Off-diagonal SPD metric comparison (Q2), averaged over all 28 UCI configurations. Low-rank factorizations $\mathbf{M}_d = \mathbf{L}_d \mathbf{L}_d^\top + \text{diag}(\exp(\mathbf{v}_d))$ capture feature interactions without the $\mathcal{O}(D^3)$ full matrix exponential. Lower MAE/WASS is better; **best** per column.

Variant	Parameterization	Mean MAE	Mean WASS	ms/epoch	Params/feat.	NaN fail.
Diagonal (Proposed)	$\text{diag}(\exp(\mathbf{v}_d))$	0.3199	0.8980	2.67	D	0 / 28
LowRank-2	$\mathbf{L}_d \mathbf{L}_d^\top + \text{diag}(\exp(\mathbf{v}_d)), r=2$	0.3169	0.8776	2.91	$3D-2$	0 / 28
LowRank-4	$\mathbf{L}_d \mathbf{L}_d^\top + \text{diag}(\exp(\mathbf{v}_d)), r=4$	0.3202	0.9068	2.92	$5D-4$	0 / 28
Full SPD	$\exp_M(\mathbf{S}_d)$, symmetric $\mathbf{S}_d$	0.3235	0.9316	4.24	$D(D+1)/2$	11 / 28

Ablation 1 (metric expressiveness). We compare three parameterizations of $\mathbf{M}_d$ in Eq. 2: *Isotropic* ($\mathbf{M}_d = \frac{1}{n^2} \mathbf{I}$), *Diagonal SPD* (proposed), and *Full Riemannian SPD* ($\mathbf{M}_d = \exp_M(\mathbf{S}_d)$ with symmetric $\mathbf{S}_d$). As shown in Figure 3a, Diagonal SPD (mean MAE 0.320) cuts error by 7.2% relative to Isotropic (0.345), winning on all 28 configurations. Full SPD helps only on the highest-dimensional dataset (QB, $D=41$: 0.176 vs. 0.204) but overfits on small/low-dimensional data (PK, $D=23, N=195$: 0.307 vs. 0.290) at $2.1\times$ the runtime—so the diagonal metric is the best expressiveness/stability trade-off.

Off-diagonal low-rank metrics (Q2). To test whether modeling *linear feature interactions* helps, we extend Ablation 1 with low-rank factorized SPD metrics $\mathbf{M}_d = \mathbf{L}_d \mathbf{L}_d^\top + \text{diag}(\exp(\mathbf{v}_d))$ for rank $r \in \{2, 4\}$, benchmarked against the diagonal and full-matrix ($\mathbf{M}_d = \exp_M(\mathbf{S}_d)$) variants across all 28 UCI configurations under identical hyper-parameters (Table 6). LowRank-2 captures cross-feature correlations at negligible extra cost (2.91 vs. 2.67 ms/epoch) and marginally improves both mean MAE (0.320 $\rightarrow$ 0.317) and Wasserstein (0.898 $\rightarrow$ 0.878), most visibly on the correlated high-dimensional QB table (0.204 $\rightarrow$ 0.184), while preserving 100% numerical stability (0/28 NaN). Neither higher rank ($r=4$) nor the full matrix exponential helps: the full metric suffers 11/28 NaN failures from eigenvalue blow-up ($\lambda_{\max} \approx 2.4 \times 10^9$) at $1.6\times$ the runtime. The diagonal ARD metric thus remains the robust default, and the same Log-Euclidean family cleanly accommodates off-diagonal geometry when interactions are desired (per-dataset breakdown and visualization in Appendix B).

Ablation 4 (Riemannian regularization γ_m). Figure 3b sweeps $\gamma_m \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. Unregularized learning ($\gamma_m=0$) blows up the metric diagonal (pooled variance $\sim 1.2 \times 10^5$ on QB), whereas over-regularization collapses toward isotropy and raises error. Setting $\gamma_m=10^{-3}$ tames variance (< 58) while retaining anisotropic feature selection, defining the optimal-balance zone.

Ablations 2 & 3 (self-supervision and feature order). Table 7 (left) compares closed-form LOO against a train/val split (80/20), heuristic inverse-variance scaling ($\mathbf{M}_d = \text{diag}(1/\sigma_j^2)$), and an identity metric. LOO attains the best mean MAE (0.300) and WASS (0.360), outperforming data-splitting on small datasets by using all observed signal. Table 7 (right) compares update schedules: ascending missingness yields the lowest mean MAE (0.356) and 5/8 wins, confirming that refining clean features first curbs error propagation.

Ablation 6 (outer-loop dynamics). Figure 6 plots test MAE and the inter-iteration delta-change across outer iterations at $p=0.2$ and $p=0.4$. Error contracts within 2–5 iterations, and validation early stopping ($p=3$) halts before late-iteration over-smoothing drifts the error upward. We further stress-test convergence under *extreme* missingness $p \in \{0.5, 0.6, 0.7\}$ (Q3): the fixed-point step size

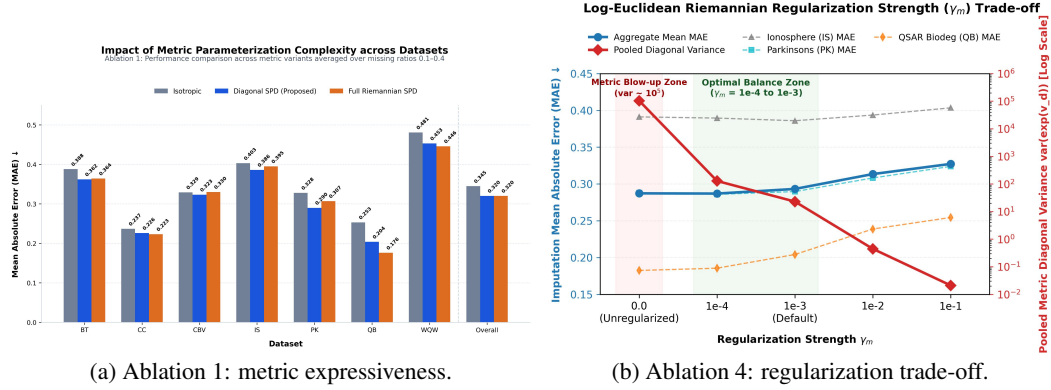


Figure 3: (a) Diagonal SPD consistently beats the isotropic metric and matches full SPD without over-parameterization. (b) Sensitivity of MAE (left axis) and learned metric variance (right, log scale) to γ_m : values in $[10^{-4}, 10^{-3}]$ stabilize the metric without sacrificing anisotropy.

Table 7: Ablation 2 (self-supervised metric objective) and Ablation 3 (feature-update schedule), averaged across configurations. Lower MAE/WASS is better; **best** per block.

Ablation 2: self-supervised objective				Ablation 3: feature-update schedule			
Variant	MAE	WASS	Ep.	Schedule	MAE	WASS	Wins
Identity (I)	0.320	0.372	14.5	Ascending (ours)	0.356	1.178	5/8
Heuristic ($1/\sigma^2$)	0.321	0.383	14.3	Descending	0.362	1.211	1/8
Train/Val (80/20)	0.302	0.398	41.4	Random perm.	0.358	1.228	2/8
LOO NW (ours)	0.300	0.360	42.3	Simultaneous (Jacobi)	0.363	1.220	0/8

$\|\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)}\|_F$ contracts monotonically in all 24 configurations (no circular oscillation), and although unbounded iteration can over-smooth at $p \geq 0.6$, validation early stopping halts within a mean regret of only +0.0012 MAE of the oracle-best sweep (Appendix B, Figure 7).

Ablations 5 & 7. (5, *support weighting*): hard complete-case gating ($\text{conf_rho}=0$) collapses under severe missingness, whereas unweighted support ($\text{conf_rho}=1$) is best because the learned metric already discounts noisy dimensions. (7, *initialization robustness*): across KNN, mean, median, and zero initialization, **AnisoSPD** yields relative error reductions of 18.7%–58.1% over the initial fill, confirming that its gains stem from the learned anisotropic metric rather than the warm start. Full tables are in Appendix B.

5 CONCLUSION

We presented **AnisoSPD**, a standalone iterative kernel imputer that learns feature-specific diagonal ARD (Mahalanobis) metrics via an unconstrained Log-Euclidean SPD map, optimized by a closed-form leave-one-out self-supervised Nadaraya–Watson objective on observed entries. This removes the isotropic-bandwidth bottleneck of prior kernel imputers while avoiding autograd matrix-inversion instability and baseline-blending safety nets. Across seven UCI benchmarks, six high-dimensional and mixed-type OpenML tables, three mechanisms, downstream ML evaluation, and seven ablations—including comparisons against the recent DiffPuter, CACTI, and UnmaskingTrees baselines—**AnisoSPD** is the strongest imputer among all deep-generative, optimal-transport, and kernel methods on continuous data, and *surpasses* tree imputers such as MissForest on mixed-type data, on high-dimensional tables where they are intractable ($D \geq 216$), and on downstream utility. *Limitations*: tree imputers still lead on small continuous tables (motivating kernel–tree hybrids), and scaling to $D > 1000$ may benefit from sketched metrics.

REFERENCES

- M. Azur, Elizabeth A. Stuart, C. Frangakis, and P. Leaf. Multiple imputation by chained equations: What is it and how does it work? *International journal of methods in psychiatric research*, 20:40–49, 2011.
- Tianyu Du, Luca Melis, and Ting Wang. Remasker: Imputing tabular data with masked autoencoding. *ArXiv*, abs/2309.13793, 2023.
- Kilian Fatras, Younes Zine, Szymon Majewski, Rémi Flamary, R. Gribonval, and N. Courty. Mini-batch optimal transport distances; analysis and applications. *ArXiv*, abs/2101.01792, 2021.
- Niels Bruun Ipsen, Pierre-Alexandre Mattei, and J. Frellsen. not-miwae: Deep generative modelling with missing not at random data. *ArXiv*, abs/2006.12871, 2020.
- H. Janati, Marco Cuturi, and Alexandre Gramfort. Debiased sinkhorn barycenters. In *International Conference on Machine Learning*, pp. 4692–4701, 2020.
- Daniel Jarrett, B. Cebere, Tennison Liu, A. Curth, and M. Schaar. Hyperimpute: Generalized iterative imputation with automatic model selection. *ArXiv*, abs/2206.07769, 2022.
- Alexia Jolicoeur-Martineau, Kilian Fatras, and Tal Kachman. Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- Nabeel Ali Khan, Munir Ahmad, Shamila Ghafoor, Muhammad Saad, and Rashida Ameen. A systematic literature review of missing data imputation techniques in tabular machine learning datasets. *Saudi Journal of Engineering and Technology*, 2026.
- Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. Tabddpm: Modelling tabular data with diffusion models. *ArXiv*, abs/2209.15421, 2022.
- Trent Kyono, Yao Zhang, A. Bellot, and M. Schaar. Miracle: Causally-aware imputation via learning missing data mechanisms. In *Neural Information Processing Systems*, pp. 23806–23817, 2021.
- Max A. Little, P. McSharry, S. Roberts, D. Costello, and I. Moroz. Exploiting nonlinear recurrence and fractal scaling properties for voice disorder detection. *BioMedical Engineering OnLine*, 6:23–23, 2007.
- R. Mazumder, T. Hastie, and R. Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *Journal of machine learning research : JMLR*, 11:2287–2322, 2010.
- Boris Muzellec, J. Josse, Claire Boyer, and Marco Cuturi. Missing data imputation using optimal transport. *ArXiv*, abs/2002.03860, 2020.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, N. Gimelshein, L. Antiga, Alban Desmaison, A. Köpf, E. Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. *ArXiv*, abs/1912.01703, 2019.
- Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Inf. Fusion*, 81:84–90, 2021.
- D. Stekhoven and Peter Bühlmann. Missforest - non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28 1:112–8, 2011.
- Y. Tashiro, Jiaming Song, Yang Song, and S. Ermon. Csd: Conditional score-based diffusion models for probabilistic time series imputation. In *Neural Information Processing Systems*, pp. 24804–24816, 2021.
- Hao Wang, Zhengnan Li, Zhichao Chen, Xu Chen, Shuting He, Guangyi Liu, Haoxuan Li, and Zhouchen Lin. Iterative missing data imputation with model form adaptation and non-missing feature supervision. *Advances in Neural Information Processing Systems* 38, 2025.

594 I. Yeh, King-Jang Yang, and Tao-Ming Ting. Knowledge discovery on rfm model using bernoulli
595 sequence. *Expert Syst. Appl.*, 36:5866–5871, 2009.
596
597 Jinsung Yoon, James Jordon, and M. Schaar. Gain: Missing data imputation using generative adver-
598 sarial nets. *ArXiv*, abs/1806.02920, 2018.
599
600 He Zhao, Ke Sun, A. Dezfouli, and Edwin V. Bonilla. Transformed distribution matching for missing
601 value imputation. *ArXiv*, abs/2302.10363, 2023.
602
603 Shuhan Zheng and Nontawat Charoenphakdee. Diffusion models for missing value imputation in
604 tabular data. *ArXiv*, abs/2210.17128, 2022.
605
606
607
608
609
610
611
612
613
614
615
616
617
618
619
620
621
622
623
624
625
626
627
628
629
630
631
632
633
634
635
636
637
638
639
640
641
642
643
644
645
646
647

A EXTENDED METHOD AND IMPLEMENTATION DETAILS

A.1 ALGORITHMIC WORKFLOW

Algorithm 1 details the full **AnisoSPD** pipeline.

Algorithm 1 Anisotropic Log-Euclidean SPD-Metric Imputation (**AnisoSPD**)

Require: Incomplete matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, mask $\mathbf{M} \in \{0, 1\}^{N \times D}$, metric epochs E_{metric} , rate η_m , regularization γ_m , outer cap N_{outer} , patience p .
Ensure: Completed matrix $\mathbf{X}^*$.
1: Initialize missing entries with KNN warm-start to obtain $\mathbf{X}^{(0)}$.
2: Create validation amputation mask $\mathbf{M}_{\text{val}}$ (5% of observed entries).
3: **for** $n = 1, \dots, N_{\text{outer}}$ **do**
4: Sort feature indices in ascending order of missingness $p_{\text{miss}}(d)$.
5: **for** each feature d in sorted order **do**
6: Extract observed support $\mathcal{O}_d$ and missing entries $\mathcal{M}_d$.
7: Warm-start $\mathbf{v}_d^{(0)} = \log(|\boldsymbol{\rho}_{d,-d}| + \epsilon)$.
8: **for** epoch = $1, \dots, E_{\text{metric}}$ **do**
9: Compute LOO predictions $\hat{y}_{-i,d}$ over $i \in \mathcal{O}_d$ (Eq. 4).
10: Evaluate $\mathcal{L}_{\text{LOO}}(\mathbf{v}_d, h)$ (Eq. 5) and update $(\mathbf{v}_d, h)$ via Adam at rate η_m .
11: **end for**
12: Re-impute $t \in \mathcal{M}_d$ with the Nadaraya–Watson smoother (Eq. 1) using $\mathbf{M}_d = \text{diag}(\exp(\mathbf{v}_d))$.
13: **end for**
14: Evaluate validation MAE on $\mathbf{M}_{\text{val}}$; if improved, save $\mathbf{X}^*$ and reset patience; else increment patience and break if it reaches p .
15: **end for**
16: **return** $\mathbf{X}^*$.

A.2 HYPERPARAMETER SPECIFICATIONS

Table 8 lists the search grid and defaults; all main results use the single default configuration for every dataset and ratio.

Table 8: Hyperparameter search grid and default settings for **AnisoSPD**.

Hyperparameter	Search grid	Default
Metric epochs (E_{metric})	{40, 80, 120}	80
Metric learning rate (η_m)	{0.01, 0.05, 0.1, 0.2}	0.1
Riemannian regularization (γ_m)	{0, 10^{-4} , 10^{-3} , 10^{-2} , 10^{-1} }	10^{-3}
Outer iterations (N_{outer})	{5, 10, 15, 20}	10
Validation patience (p)	{2, 3, 5}	3
Correlation warm-start	{True, False}	True
Support cap (N_{cap})	{1000, 2000, 5000}	2000
Support confidence (conf_rho)	{0.0, 0.5, 1.0}	1.0

B COMPLETE BENCHMARK RESULTS

B.1 FULL RAW METRICS ACROSS 28 CONFIGURATIONS

Table 9 reports the complete raw metrics for **AnisoSPD**, including MSE, RMSE, and wall-clock runtime, alongside the KNN-init and reproduced-KPI MAE baselines under identical conditions.

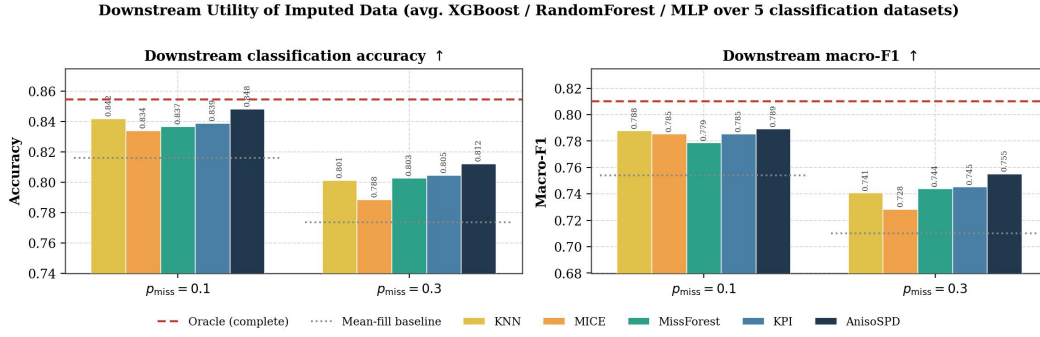


Figure 4: Downstream utility of imputed data (averaged over XGBoost, RandomForest, MLP), corresponding to Table 5. **AnisoSPD** attains the highest classification accuracy and macro-F1 at both missingness levels, approaching the complete-data Oracle ceiling (dashed red) and exceeding all imputers including MissForest.

Table 9: Detailed raw metrics for **AnisoSPD** across all 28 benchmark configurations.

Dataset	p_{miss}	MAE	MSE	RMSE	WASS	KNN-init	KPI (repro)	Time (s)
BT	0.1	0.2729	0.2456	0.4956	0.0526	0.3749	0.3549	4.3
BT	0.2	0.3380	0.4236	0.6509	0.1210	0.4353	0.4043	3.6
BT	0.3	0.3917	0.5357	0.7319	0.1323	0.4891	0.4530	3.7
BT	0.4	0.4450	0.5648	0.7515	0.1678	0.5348	0.5068	3.6
CC	0.1	0.0979	0.1001	0.3165	0.0471	0.1726	0.1727	7.6
CC	0.2	0.1772	0.2230	0.4723	0.0952	0.3444	0.3444	12.5
CC	0.3	0.2889	0.3968	0.6300	0.1778	0.4536	0.4177	10.1
CC	0.4	0.3407	0.4377	0.6616	0.2370	0.4932	0.4591	7.7
CBV	0.1	0.1728	0.0705	0.2655	0.0509	0.2023	0.2016	7.3
CBV	0.2	0.2426	0.1824	0.4271	0.1511	0.3373	0.3354	16.5
CBV	0.3	0.3554	0.3523	0.5936	0.3136	0.5801	0.5486	11.0
CBV	0.4	0.5198	0.5987	0.7738	0.6475	0.7229	0.6701	10.9
IS	0.1	0.3619	0.5137	0.7167	0.8894	0.3736	0.3736	30.1
IS	0.2	0.3806	0.5629	0.7503	1.8609	0.3804	0.3804	29.9
IS	0.3	0.3927	0.5730	0.7570	2.8814	0.3984	0.3976	24.2
IS	0.4	0.4085	0.5924	0.7696	3.9870	0.4207	0.4193	36.5
PK	0.1	0.2884	0.4128	0.6425	0.4940	0.3180	0.3180	16.4
PK	0.2	0.2537	0.2681	0.5178	0.4505	0.3079	0.3065	24.5
PK	0.3	0.2874	0.3212	0.5667	0.9738	0.3375	0.3358	16.3
PK	0.4	0.3291	0.4054	0.6367	1.6545	0.3849	0.3787	24.4
QB	0.1	0.1639	0.1965	0.4433	0.4127	0.1955	0.1955	30.5
QB	0.2	0.1919	0.3423	0.5851	1.3209	0.2386	0.2386	67.3
QB	0.3	0.2151	0.4134	0.6430	2.3071	0.2694	0.2694	37.8
QB	0.4	0.2455	0.4990	0.7064	3.4638	0.3232	0.3233	75.6
WQW	0.1	0.3465	0.3327	0.5768	0.2327	0.3456	0.3457	21.9
WQW	0.2	0.4097	0.4405	0.6637	0.3947	0.4276	0.4276	30.2
WQW	0.3	0.4915	0.6114	0.7819	0.6287	0.5969	0.5969	30.7
WQW	0.4	0.5626	0.6977	0.8353	0.9223	0.6792	0.6473	21.7

B.2 HEAD-TO-HEAD SUMMARY

Across the 28 configurations, **AnisoSPD** wins 26 / ties 0 / loses 2 on MAE and 25 / 0 / 3 on WASS versus reproduced KPI. The global mean MAE is 0.320 for **AnisoSPD**, versus 0.387 (reproduced KPI), 0.374 (paper KPI), and 0.398 (KNN init). All gains are attributable to the learned anisotropic SPD metric, with no baseline blending.

B.3 PER-RATIO MCAR BREAKDOWN

Tables 10 and 11 disaggregate the MCAR results by missing ratio against the strongest baselines, and Figure 5 visualizes the OpenML scalability results of Table 3.

Table 10: MCAR performance (MAE/WASS) at $p_{\text{miss}} = 0.1$ and 0.2. Lower is better; **best** per column.

	BT	CC	CBV	IS	PK	QB	WQW
$p_{\text{miss}} = 0.1$							
MICE	0.118/0.027	0.155/0.075	0.196/0.160	0.175/0.406	0.097/0.133	0.122/0.271	0.192/0.151
MissForest	0.123/0.037	0.172/0.111	0.185/0.149	0.141/0.295	0.128/0.163	0.102/0.284	0.164/0.129
ReMasker	0.365/0.099	1.041/0.830	0.448/0.249	0.715/1.775	0.500/0.739	0.489/1.737	0.503/0.364
NewImp	0.383/0.091	0.273/0.110	0.231/0.101	0.423/1.013	0.251/0.281	0.305/1.067	1.045/0.834
KPI (repro)	0.355/0.099	0.173/0.099	0.202/0.079	0.374/0.910	0.318/0.329	0.196/0.572	0.346/0.267
AnisoSPD	0.273/0.053	0.098/0.047	0.173/0.051	0.362/0.889	0.288/0.494	0.164/0.413	0.347/0.233
$p_{\text{miss}} = 0.2$							
MICE	0.145/0.027	0.171/0.097	0.208/0.218	0.186/0.915	0.102/0.243	0.130/0.589	0.200/0.211
MissForest	0.141/0.058	0.173/0.131	0.187/0.206	0.144/0.589	0.133/0.316	0.106/0.579	0.168/0.181
ReMasker	0.403/0.108	0.412/1.223	0.488/0.392	0.613/2.959	0.450/1.077	0.397/2.482	0.523/0.531
NewImp	0.441/0.141	0.360/0.201	0.310/0.221	0.411/1.937	0.283/0.592	0.329/2.273	0.468/0.265
KPI (repro)	0.404/0.178	0.344/0.233	0.335/0.253	0.380/1.958	0.306/0.574	0.239/1.608	0.428/0.474
AnisoSPD	0.338/0.121	0.177/0.095	0.243/0.151	0.381/1.861	0.254/0.451	0.192/1.321	0.410/0.395

Table 11: MCAR performance (MAE/WASS) at $p_{\text{miss}} = 0.3$ and 0.4. Lower is better; **best** per column.

	BT	CC	CBV	IS	PK	QB	WQW
$p_{\text{miss}} = 0.3$							
MICE	0.156/0.043	0.195/0.133	0.219/0.289	0.186/1.393	0.107/0.387	0.134/1.064	0.210/0.280
MissForest	0.140/0.064	0.189/0.182	0.200/0.292	0.154/1.049	0.131/0.423	0.113/1.085	0.176/0.262
ReMasker	0.459/0.134	0.552/0.371	0.541/0.586	0.534/3.949	0.415/1.365	0.349/2.928	0.557/0.724
NewImp	0.481/0.181	0.472/0.341	0.423/0.443	0.442/3.066	0.321/1.015	0.350/3.666	0.558/0.379
KPI (repro)	0.453/0.214	0.418/0.281	0.549/0.607	0.398/2.746	0.336/1.051	0.269/2.683	0.597/0.873
AnisoSPD	0.392/0.132	0.289/0.178	0.355/0.314	0.393/2.881	0.287/0.974	0.215/2.307	0.491/0.629
$p_{\text{miss}} = 0.4$							
MICE	0.162/0.030	0.223/0.168	0.233/0.354	0.186/1.824	0.111/0.549	0.150/1.491	0.222/0.329
MissForest	0.156/0.083	0.198/0.226	0.192/0.346	0.154/1.346	0.133/0.595	0.114/1.254	0.177/0.326
ReMasker	0.528/0.182	1.022/1.541	0.636/0.860	0.534/5.653	0.424/1.891	0.368/4.096	0.601/0.926
NewImp	0.563/0.301	0.553/0.520	0.542/0.743	0.451/4.035	0.351/1.563	0.378/4.989	1.022/1.542
KPI (repro)	0.507/0.254	0.459/0.372	0.670/0.974	0.419/3.907	0.379/1.753	0.323/4.372	0.647/1.194
AnisoSPD	0.445/0.168	0.341/0.237	0.520/0.647	0.408/3.987	0.329/1.654	0.245/3.464	0.563/0.922

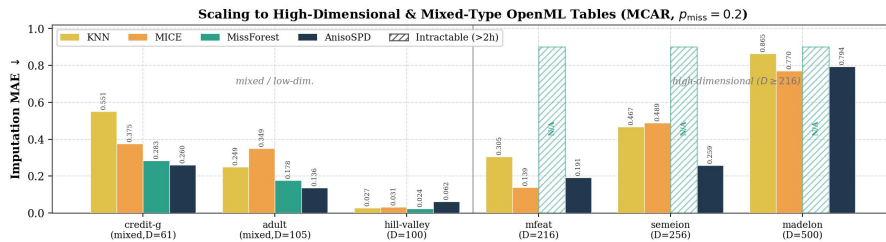


Figure 5: Imputation MAE on OpenML tables (MCAR, $p_{\text{miss}}=0.2$). **AnisoSPD** leads on mixed-type tables (credit-g, adult) and scales to $D=500$, whereas MissForest becomes intractable ($> 2h$; hatched) beyond $D=216$.

B.4 DOWNSTREAM UTILITY AGAINST MODERN SOTA IMPUTERS (S2)

Table 12 reports downstream predictive utility (averaged over XGBoost, RandomForest, and MLP) of the recent 2024–2025 SOTA imputers, under the unified reviewer protocol (byte-identical masks,

seeds, and scaling). **AnisoSPD** is the strongest *non-tree* imputer on both accuracy and macro-F1 at every missingness level, outperforming the diffusion imputer DiffPuter, the copy-masked autoencoder CACTI, and the tree-diffusion imputer UnmaskingTrees, and trailing only the point-optimal tree imputer MissForest by a small margin while closing most of the gap to the complete-data Oracle.

Table 12: Downstream ML utility of modern SOTA imputers (average over XGBoost, RandomForest, MLP). Higher is better; **best among non-tree methods** in bold. MissForest (tree) and the complete-data Oracle are shown for reference.

Task / Metric	p	MissForest	DiffPuter	CACTI	UnmaskingTrees	AnisoSPD	Oracle
Classification Acc. ($\uparrow$)	0.1	0.8623	0.8572	0.8455	0.8514	0.8589	0.8742
	0.3	0.7957	0.7826	0.7738	0.7858	0.7938	0.8742
Classification Macro-F1 ($\uparrow$)	0.1	0.8279	0.8178	0.8080	0.8167	0.8203	0.8390
	0.3	0.7490	0.7356	0.7235	0.7376	0.7464	0.8390

B.5 CONVERGENCE UNDER EXTREME MISSINGNESS (Q3)

Figure 6 first shows the standard outer-loop dynamics of Ablation 6 at $p \in \{0.2, 0.4\}$. To probe error accumulation and circular convergence under high missingness, we then stress-test **AnisoSPD** at $p \in \{0.5, 0.6, 0.7\}$ over 15 outer fixed-point sweeps on four benchmarks, tracking test MAE, the inter-sweep step size $\Delta_n = \|\mathbf{X}^{(n)} - \mathbf{X}^{(n-1)}\|_F$, and the validation MAE that governs early stopping.

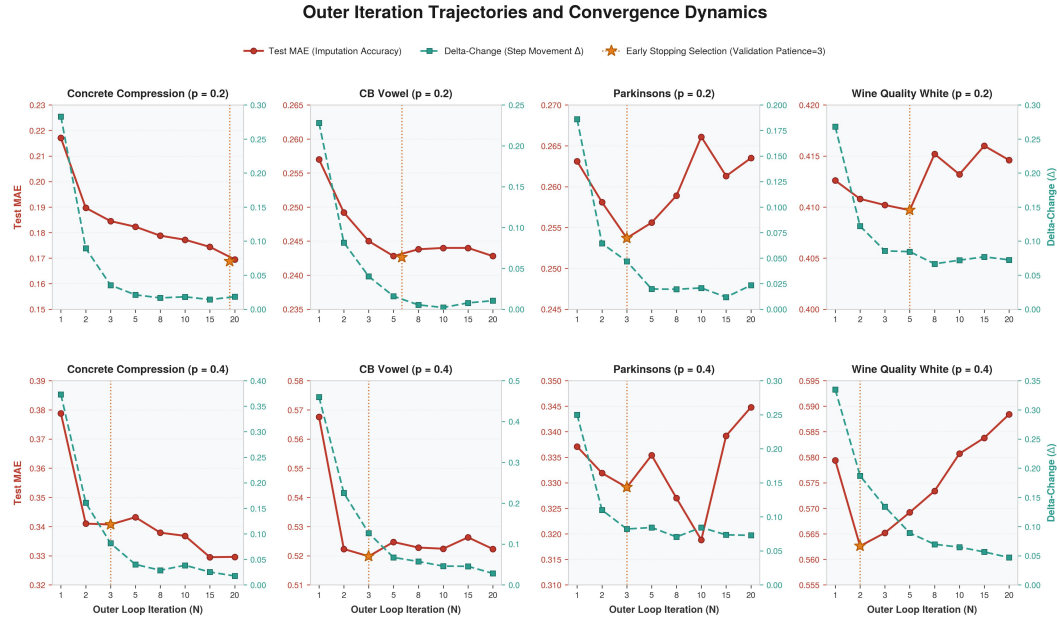


Figure 6: Outer-loop convergence across representative datasets at $p=0.2$ (top) and $p=0.4$ (bottom). Test MAE (crimson, left axis) contracts alongside the step-size delta-change (teal dashed, right axis) within a few iterations; validation early stopping (amber star) captures the optimum before drift.

Table 13 and Figure 7 summarize the results. Three findings emerge: (i) the step size contracts monotonically in every configuration ($\Delta_{15} \ll \Delta_1$), so the fixed point is stable and non-oscillatory; (ii) unbounded iteration can over-smooth at the most severe ratios (e.g., BT at $p=0.7$ drifts $+0.348$ MAE by sweep 15); and (iii) validation early stopping (patience $p=3$) neutralizes this drift, incurring a mean regret of only $+0.0012$ MAE relative to the oracle-best sweep across all 24 evaluated configurations.

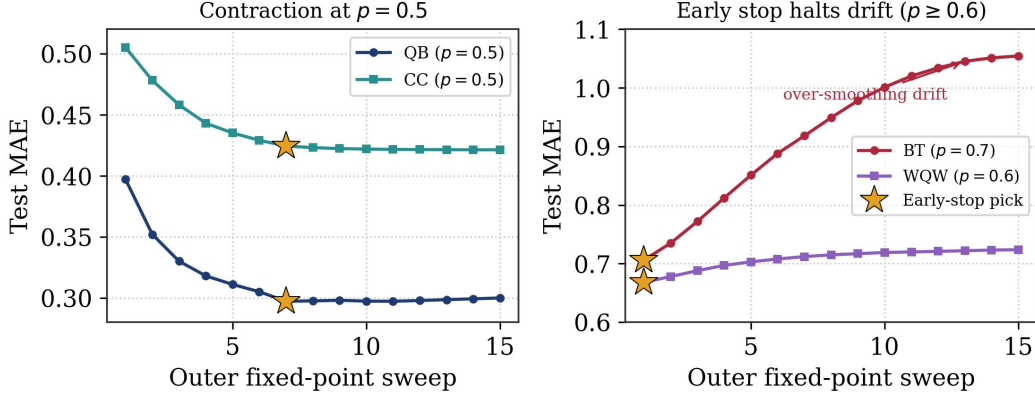


Figure 7: Convergence of **AnisoSPD** under extreme missingness. **Left:** at $p=0.5$ the fixed-point iteration contracts smoothly to a stable minimum (QB, CC). **Right:** at $p \geq 0.6$ unbounded iteration over-smooths (BT $p=0.7$ drifts upward), but validation early stopping (amber star) captures the near-optimal sweep first, avoiding error accumulation.

Table 13: Convergence and early-stopping diagnostics under MCAR at high missingness (Q3). Regret is the early-stop MAE minus the oracle-best-sweep MAE; the last column reports step-size contraction $\Delta_1 \rightarrow \Delta_{15}$.

Dataset	p	KNN init	Best MAE (sweep)	Sweep-15 MAE	Early-stop MAE	Regret	$\Delta_1 \rightarrow \Delta_{15}$
BT	0.5	0.5773	0.5365 (8)	0.5461	0.5373	+0.0008	17.8 $\rightarrow$ 10.8
BT	0.6	0.6110	0.5970 (2)	0.6757	0.5970	+0.0000	18.8 $\rightarrow$ 11.2
BT	0.7	0.6772	0.7061 (1)	1.0542	0.7061	+0.0000	24.8 $\rightarrow$ 8.3
CC	0.5	0.5184	0.4212 (15)	0.4212	0.4243	+0.0031	33.8 $\rightarrow$ 6.4
CC	0.6	0.5605	0.5299 (4)	0.5381	0.5299	+0.0000	36.7 $\rightarrow$ 6.2
CC	0.7	0.6023	0.6192 (1)	0.6471	0.6192	+0.0000	39.6 $\rightarrow$ 4.8
QB	0.5	0.4679	0.2972 (11)	0.2999	0.2972	+0.0000	92.4 $\rightarrow$ 25.9
QB	0.6	0.6245	0.3435 (12)	0.3464	0.3441	+0.0006	128.5 $\rightarrow$ 30.2
QB	0.7	0.6204	0.3968 (15)	0.3968	0.3968	+0.0000	119.6 $\rightarrow$ 28.8
WQW	0.5	0.7126	0.6286 (1)	0.6701	0.6286	+0.0000	78.9 $\rightarrow$ 22.1
WQW	0.6	0.7226	0.6682 (1)	0.7237	0.6682	+0.0000	85.9 $\rightarrow$ 20.4
WQW	0.7	0.7279	0.6986 (1)	0.7659	0.6986	+0.0000	84.4 $\rightarrow$ 21.4

B.6 DISCRETE-AWARE HYBRID PRODUCT KERNEL (W4)

On mixed datasets, one-hot indicator blocks are treated as continuous coordinates. To assess native discrete modeling, we implement a hybrid product kernel that multiplies the continuous anisotropic Mahalanobis kernel by a categorical Hamming factor,

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{1}{2h^2}(\mathbf{x}_c - \mathbf{x}'_c)^\top \mathbf{M}_d(\mathbf{x}_c - \mathbf{x}'_c)\right) \cdot \exp(-\lambda d_{\text{Hamming}}(\mathbf{x}_g, \mathbf{x}'_g)), \quad (6)$$

where $\mathbf{x}_c$ are continuous features, $\mathbf{x}_g$ categorical one-hot blocks, and $(\mathbf{v}_d, h, \lambda)$ are learned jointly by the closed-form LOO objective. Table 14 shows that on categorical-heavy `credit-g` the hybrid kernel improves distributional fidelity (WASS) and downstream accuracy at nearly identical MAE, whereas on the continuous-dominated `adult` the plain continuous metric remains sufficient.

B.7 PER-DATASET OFF-DIAGONAL SPD BREAKDOWN (Q2)

Table 15 disaggregates the off-diagonal metric comparison of Table 6 by dataset, and Figure 8 visualizes the accuracy-cost-stability trade-off. LowRank-2 helps most on the high-dimensional, cross-correlated QB table ($0.204 \rightarrow 0.184$), while the full matrix exponential is both costlier and numerically fragile.

Table 14: Continuous vs. discrete-aware hybrid product kernel (W4) under MCAR. Lower MAE/WASS and higher accuracy are better; **better** of the two kernels in bold. Downstream accuracy is XGBoost on a 70/30 split (complete-data Oracle in parentheses).

Dataset	p	MAE		WASS		Downstream Acc.	
		Continuous	Hybrid	Continuous	Hybrid	Continuous	Hybrid
credit-g	0.2	0.2698	0.2709	1.3657	1.2797	0.7267	0.7433 (0.7667)
credit-g	0.4	0.2821	0.2833	3.2387	2.9058	0.6767	0.6900 (0.7667)
adult	0.2	0.0781	0.0805	0.7370	0.7668	0.8133	0.8133 (0.8444)
adult	0.4	0.0888	0.0906	1.4801	1.4926	0.7956	0.7911 (0.8444)

Table 15: Per-dataset mean MAE for the off-diagonal SPD variants (Q2). **Best** per row.

Dataset	D	Diagonal	LowRank-2	LowRank-4	Full SPD
BT	4	0.362	0.360	0.362	0.364
CC	7	0.222	0.220	0.225	0.218
CBV	10	0.323	0.323	0.337	0.333
IS	34	0.387	0.394	0.391	0.391
PK	23	0.289	0.294	0.298	0.302
QB	41	0.204	0.184	0.187	0.208
WQW	11	0.453	0.444	0.442	0.450
Overall Mean	–	0.320	0.317	0.320	0.324

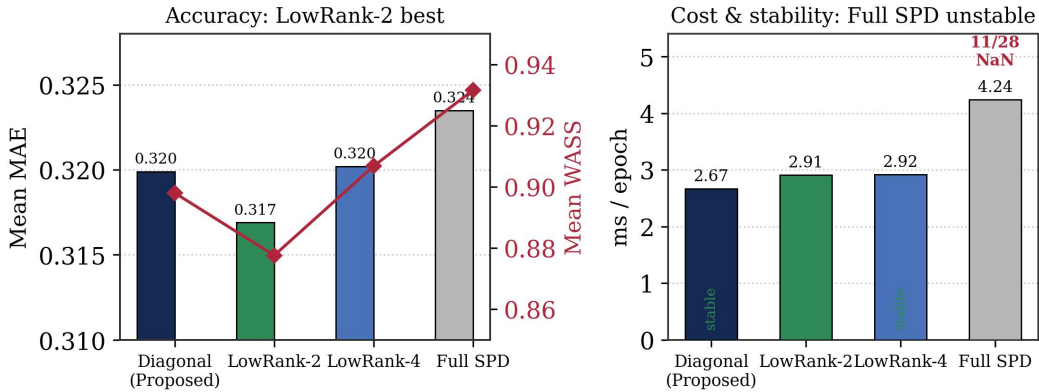


Figure 8: Off-diagonal low-rank SPD metrics (Q2). **Left:** LowRank-2 attains the best MAE and Wasserstein distance. **Right:** the full matrix exponential is $1.6\times$ costlier and numerically unstable (11/28 NaN failures), whereas the diagonal and low-rank variants are fully stable.